7.4 What Can I Do When I Know About Prime Numbers?

1. Word Problem: Aly has a garden with a fence on 3 sides of it. There are 11 feet of fencing and her garden is 14 square feet. What is the length of the shorter side of her garden?

   2 feet

2. Cedric has rectangular handkerchief that is 21 square inches and has 20 inches of border. What is the length of the shorter side of the rectangle?

   3 inches

3. The Weasley Family has 9 members. If they want to divide up into multiple teams of more than one player each to play a game, can they do this? What if the two parents don't play?

   Yes (3 teams of 3). No if the parents don't play.

4. Making Connections: Many times when we measure angles, we say that 360 degrees is all the way around a circle. This was decided because 360 has many factors, and so you can divide it into 2, 3, 4, 5, 6, 8, 9, 10, and many other numbers of angles with the angle still having an integer degree measure. If one were to create a new system of measuring angles, which of the following numbers of degrees would be best? Which would be worst? Why?

   111 120 122 124 132 137

   120 is probably best – it can be divided by 2, 3, 5, 6, 8, 10, etc.

   137 is probably worst. It is prime.

5. Challenge: Molly has a rectangular garden with an area of 175 square feet. Both the length and the width (in feet) are whole numbers. She wants to put a fence along all four sides of the garden. What are all the possibilities for the length of fencing she might need?

   64 feet (7 × 25), 80 feet (5 × 35), 352 feet (1 × 175)

Notes on 1: The only ways to get an area of 14 square feet with integer side lengths is with a 2 by 7 or 1 by 14 garden. Here, 2 by 7 works – the fencing is along both short sides and one long side. On 2: The rectangle is 3 inches by 7 inches. On 5: It helps to think about the prime factorization of $175 = 5 \times 5 \times 7$. The shorter side will need to be 1, 5 or 7 feet. The longer side will need to be a product of the other two numbers (or all three if the short side is one foot.)

7.5 Which Numbers Are Prime?

1. Word problem: Ella is having a birthday party. She can invite as few as 10 people or as many as 50. However, all of her friends like to play a game which requires that there be multiple teams of at least 2 people. How many numbers of people can she invite?

   $\boxed{30}$

2. How many 1 or 2 digit primes end in a 0? ___0___ a 1?___5___ a 2?___1___ a 3?___7___ a 4?___0___ a 5?___1___ a 6?___0___ a 7?___6___ an 8? ___0___ a 9?___5___

3. What is the smallest prime bigger than 17?

   $\boxed{19}$

4. How many primes are between 40 and 50?

   $\boxed{3}$

5. The smallest prime number p for which p+10 and p+20 are also prime is 3. What is the smallest prime number p for which p + 30 and p + 60 are also prime?

   $\boxed{7}$

6. Making Connections: What prime is closest to 367 – 242 + 33 – 58 + 4 – 10?

   $\boxed{97}$

7. Challenge: The first five primes – 2, 3, 5, 7, and 11 – are all palindromes: they stay the same if you reverse their digits. After this, there are N primes that are not palindromes before you get to the next prime palindrome. What is N?

   $\boxed{20}$

Notes on 1: She can pick any composite number from 10 to 50. There are 41 numbers from 10 to 50 and 11 of them are prime: 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47. On 6: The connection is to section 3.2. The additions and subtractions are easier if you reorder as (367+33) – (242+58) + 4 – 10 = 400 – 300 + 4 – 10 = 94. On 7: All two digit palindromes after 11 are not prime (because 22, 33, … are all multiples of 11). The first three digit palindrome, 101, is prime. So the answer to the questions is just the number of primes between 11 and 101. There are 20 of them.

7.6 Twin Primes

1. Word problem: Em is exactly two years younger than Henry. If Em is 42 when she and Henry go to Oz and stop aging, how many times will their ages have both been prime numbers at the same time?

$\boxed{6}$

2. List all the pairs of twin primes less than 100.

$\boxed{(3,5),\ (5,7),\ (11,13),\ (17,19),\ (29,31),\ (41,43),\ (59,61),\ (71,73)}$

3. Cousin primes are primes that differ by 4. How many pairs of cousin primes are there where both numbers are less than 100?

$\boxed{8}$

(They are (3,7), (7,11), (13,17), (19,23), (37,41), (43,47), (67,71), and (79,83).)

4. Challenge: What is the largest pair of twin primes smaller than 300? What is the smallest pair of twin primes larger than 300?

$\boxed{(281, 283) \text{ and } (311, 313)}$

Notes on 1: The question is asking for the number of pairs of twin primes with the smaller number at most 42. There are 6 such pairs: (3, 5), (5, 7), (11, 13), (17, 19), (29, 31), and (41, 43). On 4: A good way to work on these problems is to think about the pattern of where in every 30 numbers you might find twin primes: the numbers that are (11, 13), (17, 19), and (29, 31) more than a multiple of 30. 270 is a multiple of 30 so the three pairs of numbers that might be twin primes in the 280's, 290's, and 300's are (281, 283), (287, 289), and (299, 301). 289 is not prime (it's 17 × 17) so the largest pair less than 300 is (281, 283). To look for a pair greater than 300 note that in the 310's, 320's, and 330's the possible twin primes are (311, 313), (317, 319), and (329, 331). The first pair does work.

8.1 Subtraction

1. Word Problem: If Ender led 94 ships into battle against aliens, and 82 were destroyed, how many ships were not destroyed?

   $\boxed{12}$

2. Rewrite the following problems with one number on top of the other and subtract.

   234 – 123 = 111 563 – 421 = $\boxed{142}$ 829 – 17 = $\boxed{812}$

   $$\begin{array}{r} 234 \\ \underline{-123} \\ 111 \end{array}$$

   777 – 673 = $\boxed{104}$ 8243 – 31 = $\boxed{8212}$ 9439 – 1029 = $\boxed{8410}$

3. Rewrite the following problems with one number on top of the other and subtract. Don't forget to line things up on the decimal points.

   124.23 – 12.21 = 112.02 894142 – 71031 = $\boxed{823{,}111}$

   $$\begin{array}{r} 124.23 \\ \underline{-12.21} \\ 112.02 \end{array}$$

   938.32 – 834.2 = $\boxed{104.12}$ 5589675.974 – 172445.22 =

   $\boxed{5{,}417{,}230.754}$

4. Making Connections: An interior angle of a regular dodecagon measures 150°. An interior angle of a regular decagon is 144°. An interior angle of a regular hexagon is 120°. Which is larger: the angle that you still need to fill in to make 360° after you put two decagons together or the angle you still need to fill in when you put a hexagon next to a dodecagon? How much bigger is it?

   $\boxed{\text{The angle to fill in next to the hexagon and dodecagon is 18° bigger.}}$

Notes on 4: The connection is to sections 4.4 and 4.5 (but nothing from those sections is needed to do the problem). The question asks which is bigger 360 – (150 + 120) or 360 – (144 + 144). You can find this by just doing the calculations. But it's quicker if you just realize that 150 + 120 is smaller than 144 + 144 so 360 – (150 + 120) is larger.

8.2 Subtraction with Borrowing

1. Word Problem: In the Olympics, Shawn Johnson did a floor routine with a maximum score of 16.3, but she made a big mistake and the judges deducted 1.4 from her maximum score. What was her final score?

   $\boxed{14.9}$

2. Subtract the following.

   2954 – 1993 = 961   271 – 139 = $\boxed{132}$ 452 – 271 = $\boxed{181}$

   $$\begin{array}{r} \scriptstyle 1\ 18\ 15 \\ \cancel{29}54 \\ \underline{-1993} \\ 961 \end{array}$$

   5621 – 39 = $\boxed{5582}$ 56554 – 5655 = $\boxed{50,899}$ 2101 – 1872 = $\boxed{229}$

   10002 – 9384 = $\boxed{618}$ 4510044 – 3625931 = $\boxed{884,113}$

3. Making Connections: Subtract the sum of all positive multiples of 7 less than 100 from the sum of all positive multiples of 5 less than 100.

   $\boxed{215}$

4. Challenge: On November 10, 2084 the Earth will pass directly between Mars and the Sun. If there are people on Mars (or any Martians there) they will see the Earth and the Moon as black dots blocking out a small part of the Sun. On that date the Earth will be 148,153,060 kilometers from the Sun. Mars will be something like 227,900,000 kilometers away from the Sun. If Mars were that far away from the Sun on that date, how far would Earth be from Mars?

   $\boxed{\text{79,746,940 kilometers}}$

Notes on 4: The connection is to section 3.3. The sums 7 + 14 + … + 98 and 5 + 10 + … + 95 can both be computed by adding pairs of numbers from opposite ends. The first is 105 × 7 = 735 and second is 9 × 100 + 50 = 950. Then 950 – 735 = 215. On 5: When the Earth is directly between Mars and the Sun you can find the Mars to Earth distance just by subtracting 227,900,000 – 148,153,060. Doing the subtraction is a good way to see if students understand how to do borrowing when you have to borrow in multiple columns in a row.

8.3 Subtracting in Your Head

1. Word Problem: Wanda got 249 pounds of flour on one particular excursion. If she and her friends then used 124 pounds, how much flour was left?

   $\boxed{\text{125 pounds}}$

2. Suppose you are doing the subtraction problem 671 – 253 in your head by first subtracting the 200, then subtracting the 50, then subtracting 3. To help keep track of things, you say the answer to yourself after each subtraction. Write down the list of numbers you say to yourself in order.

   $\boxed{471, 421, 418}$

3. Subtract the following in your head.

   295 – 126 = 169 844 – 411 = $\boxed{433}$ 452 – 248 = $\boxed{204}$

   512 – 256 = $\boxed{256}$ 565 –393 = $\boxed{172}$ 2754 – 1522 = $\boxed{1232}$

   3552 – 1334 = $\boxed{2218}$ 45,400 – 33,259 = $\boxed{12{,}141}$

4. Making Connections: Arjun puts one edge of a regular decagon (which has a 144° angle) up against the edge of a regular pentagon (which has a 108° angle) with the same edge-length. Is there a regular polygon that would exactly fit the remaining space where the angles come together?

   $\boxed{\text{Yes. A regular pentagon will fit.}}$

5. Challenge: Eunice made a big number by typing the digits in the order they appear along the top of a typewriter keyboard: 1234567890. She then made another big number by writing the numbers 13 through 17 next to each other with no spaces. And then she subtracted the smaller number from the larger one. Figure out the answer she got in your head.

   $\boxed{79{,}583{,}727}$

Notes on 4: The connection is to sections 4.4 and 4.5. The two angles add up to 252 degrees. 360 – 252 = 108. This is the angle in a regular pentagon. On 5: The question asks you to compute 1314151617 – 1234567890. There's no easy way to do this. One approach is to use subtraction with borrowing and go right to left. Another is to go one digit at a time from left to right saying the number at each step: 1314151617, 314151617, 114151617, … . The way I find easiest is to break it up after 4 digits because of the similarity in how the numbers start: 1,314,000,000 – 1,234,000,000 + 151617 – 567890. The first term is 80,000,000. But the second one is negative so I think of the answer as 79,000,000 + 151,617 + 1,000,000 – 567,890 = 79,0000,000 + 151,617 + 432,110 = 79,583,727.